Find-S Algorithm using Enjoy Sport Dataset

Project Overview

This project demonstrates the implementation of the Find-S Algorithm, one of the basic concept learning algorithms used in Machine Learning. The algorithm identifies the most specific hypothesis that fits all positive training examples from the given dataset.

The dataset used is the classic Enjoy Sport dataset, which contains weather-related attributes and whether a person enjoys sport under those conditions.

AIM

To implement the Find-S Algorithm and determine the most specific hypothesis from the Enjoy Sport dataset.

OBJECTIVE

- To understand the working of the Find-S Algorithm.
- To train the model using positive examples from the dataset.
- To generate the final hypothesis that correctly classifies the positive training instances.
- To observe how the hypothesis changes after processing each example.

DATASET USED

The following Enjoy Sport dataset is used:

Sky	Air Temp	Humidity	Wind	Water	Forecast	Enjoy Sport
Sunny	Warm	Normal	Strong	Warm	Same	Yes
Sunny	Warm	High	Strong	Warm	Same	Yes
Rainy	Cold	High	Strong	Warm	Change	No
Sunny	Warm	High	Strong	Cool	Change	Yes

Dataset Attributes

1. Sky
2. Air Temperature
3. Humidity
4. Wind
5. Water

6. Forecast
7. Target Class: Enjoy Sport

ALGORITHM DESCRIPTION

The Find-S Algorithm works only with positive examples.

1. Initialize the hypothesis with the most specific possible values.
2. Read each training example.
3. If the example is positive:
    - Compare it with the current hypothesis.
    - If an attribute differs, replace that attribute with ?.
4. Ignore negative examples.
5. The final hypothesis is the most specific rule that covers all positive examples.

FLOW OF EXECUTION

Start

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Read Dataset

↓

Initialize Hypothesis = [0,0,0,0,0,0]

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For each training example

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If example is positive

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Update hypothesis

↓

Else ignore

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Print Final Hypothesis

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Stop

PYTHON IMPLEMENTATION

```
import csv


with open('enjoy_sport.csv', 'r') as file:
    data = list(csv.reader(file))


concepts = [row[:-1] for row in data[1:]]
target = [row[-1] for row in data[1:]]


hypothesis = ['0'] * len(concepts[0])


for i in range(len(concepts)):
    if target[i].lower() == 'yes':


        if hypothesis == ['0'] * len(concepts[0]):
            hypothesis = concepts[i].copy()


        else:
            for j in range(len(hypothesis)):
                if hypothesis[j] != concepts[i][j]:
                    hypothesis[j] = '?'


print("Final Hypothesis:")
print(hypothesis)
```

HOW TO RUN

1. Save the dataset in a file named enjoy_sport.csv.
2. Save the Python program in a file named find_s.py.
3. Open terminal or command prompt.
4. Navigate to the folder containing both files.

5. Run the program using:

python find_s.py

SAMPLE OUTPUT

Final Hypothesis:

['sunny', 'warm', '?', 'strong', '?', '?']

OBSERVATIONS

- The algorithm considers only positive training examples.
- Negative examples are ignored completely.
- The hypothesis becomes more general whenever there is a mismatch between positive examples.
- The attributes Humidity, Water, and Forecast changed to ? because they had different values in positive examples.
- The attributes Sky, Air Temp, and Wind remained fixed because they were common in all positive examples.

EXPECTED OUTCOME

After executing the program, the Find-S Algorithm should generate the following final hypothesis:

['sunny', 'warm', '?', 'strong', '?', '?']

This means a person enjoys sport when:

- Sky is Sunny
- Air Temperature is Warm
- Wind is Strong
- Humidity can be anything
- Water can be anything
- Forecast can be anything

ADVANTAGES

- Simple and easy to implement.
- Works efficiently for small datasets.
- Helps understand the concept of hypothesis learning.

LIMITATIONS

- Uses only positive examples.
- Cannot handle noisy data.
- May not work well for large or complex datasets.
- Does not generate all possible hypotheses.

TOOLS AND TECHNOLOGIES

- Python 3.x
- CSV File Handling
- Basic Machine Learning Concept Learning

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